# Charles Babbage Throws a Party

## And Charles Darwin Chats Up the Ladies

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*This is basically all true. My research is based on Janet Browne's two-volume biography of Charles Darwin, my own (with Paul Freiberger)* Fire in the Valley, *and other sources. I can't definitively confirm that Darwin was present at Charles Babbage's soirée on this particular day, but he was a frequenter of Babbage's gathering during this period, and he attended at least in part for the purpose of chatting up the ladies. That he spoke with Ada Lovelace at the punchbowl is my (plausible) invention, though the punchbowl was real.*

*Every other detail — Jack the Ripper, Queen Victoria, Karl Marx, the invention of the cowcatcher, the hooligan street musicians, Charles Dickens, the tarts and cold salmon and oysters and Darwin's stomach problems, and Lord Byron's computer-savvy daughter— is historical fact.*

Greetings. We are in the heart of London at the dawn of the Victorian age — literally the dawn of the age, as the teen-aged Victoria has only recently been crowned — at a Saturday afternoon soirée at 1 Dorset Street.

## Not that Dorset Street

You may possibly recognize the name of the street. Dorset Street, in the East End of London, was known as "the worst street in London," and not only because it was the scene of the brutal murder of Mary Jane Kelly by Jack the Ripper. Dorset Street could be fairly described, using Thomas Hobbes's description of life outside civilization, as "poor, nasty, brutish, and short."

Jack London referred to its residents as "the people of the abyss," and his journalist friend Frederick Arthur McKenzie said that it "boasts of an attempt at murder on an average once a month, of a murder in every house, and one house at least, a murder in every room. Policemen go down it as a rule in pairs. Hunger walks prowling in its alleyways, and the criminals of to-morrow are being bred there to-day."

But those events are all in the future, and indeed in a different Dorset Street. Our 1 Dorset Street is a handsome home in the fashionable Marlybone section of London's West End, and in the year 1838, it is the residence of eccentric genius Charles Babbage.

Both adjectives apply. Babbage is an inventor, a mathematician, an author, a politician — and he is also a character. He holds Newton's chair in mathematics at Cambridge. He has written a book on economics that will influence Karl Marx and John Stuart Mill. He designs calculating machines. He invented the cow-catcher.

And he writes angry letters to the *Times*.

## The Curse of the Street Musicians

Musicians in 19th Century London hit upon a novel business model, one that

Midwest farmers in the U.S. would find useful in the 20th Century: getting paid not to play. Street musicians would gather outside restaurants and other establishments and start playing as loudly as they could. Typically, the proprietor would pay them something to go away and stop bothering the patrons.

They also played outside 1 Dorset Street, and Babbage was not disposed to submit to extortion. He wrote letters to the *Times*. He went to court. He enlisted his friends in his battle against street musicians.

One of these friends was the author Charles Dickens. Dickens described the strategy of the musicians: "No sooner does it become known to the producers of horrible sounds that any of your correspondents have particular need of quiet in their own houses, than the said houses are beleaguered by discordant hosts seeking to be bought off."



That should have been a tip-off to Babbage to keep a low profile and not make a target of himself. But that was not Babbage's style. He kept up his campaign and soon he had a brass band playing outside his house for hours on end and whenever he left the house he was followed and taunted. He even wrote a book about the curse of the street musicians, and that finally got some regulations passed that eased his pain.

So that was the kind of man he was, but at the moment the salient fact about Charles Babbage is that he hosts the most fashionable parties in London. And we have been invited to this particular soirée.

## The Naturalist at the Punchbowl

Babbage's soirées are an essential part of the London social scene, and invitations are highly sought after. Here, socialites and politicians mingle with authors, actors, and scientists to share the latest gossip and literary and scientific novelties. If you want to listen in on a member of young Queen Victoria's court debating the benefits of industrialization with Charles Dickens, Babbage's parlor is the place to be.

This particular Saturday afternoon features music and dancing — and much talk. Tables have been laid with tarts and finger sandwiches, oysters and cold salmon — and the drinks: wine, cordials, Madeira, punch. We find young Charles Darwin chatting up an attractive young woman by the punchbowl.

Darwin has taken up temporary residence in London to do research on his book. He is unmarried and out of touch with the London social scene, having been off on a five-year sea voyage as the naturalist aboard the HMS Beagle. The book is about that voyage. His friend Charles Lyell encourages him to start attending Babbage's soirées, getting his attention when Darwin hears that there are always attractive young women in attendance.
Back at the punchbowl, Darwin refuses an oyster, pleading chronic indigestion, exacerbated by his travels. He turns at the sound of Babbage's voice.

The highlight of the afternoon has arrived: a scientific presentation by Babbage himself. Everyone here is familiar with Babbage's Difference Engine, which he has been laboring over but still hasn't finished. The Difference Engine is a calculating machine. It is designed to perform a specific laborious mathematical task: the calculation of polynomial functions — a calculation that is critical to much scientific research. It hints at even more: it isn't hard to imagine how other, similar machines could be built to perform other number-crunching chores. The Difference Engine is the most ambitious of Babbage's inventions to date, and he is frustrated by his inability to get funding to complete it. His guests know all that. They've been here before.

But Babbage isn't interested in talking about his Difference Engine this afternoon. All his energy is now directed toward his new creation, the Analytical Engine. While the Difference Engine can perform one task, the Analytical Engine, he tells them, is a universal device. It can perform any describable computation. No need to build other calculating machines: this one is all you need or will ever need. Oh, and it's not limited to working with numbers: it can operate just as well on text or on logical propositions. A translating machine? A thinking machine? Yes, it's all of that.

## The Poet's Daughter

Some of the guests are clearly having trouble following this. One young listener, though, is hanging on Babbage's words with more interest — and probably a deeper understanding — than anyone else present. For four years now, she has been — what? Babbage's pupil, his protégé, his booster? All of that. She is Augusta Ada Byron, Countess of Lovelace, Ada to her friends, the daughter of a poet and a mathematician. While her father, Lord Byron, rebelled against society and her mother, Anne Isabella Milbanke Byron, rebelled against the idea that a woman could not be a mathematician, Ada rebelled at being defined by either of her parents. She was determined to explore both her artistic and mathematical talents. Both are in play as she listens to Babbage.

Ada understands what the other guests seem not to get: she sees how fundamentally different the Analytical Engine is. It is a device with no designed purpose. And she understands the implication of this. Yes, it can perform any task you can tell it to do, but you have to tell it, and this telling requires another new innovation.

The Difference Engine can be set to work by flipping some levers. It is a machine, and acts like one. But controlling a universal device like the Analytical Engine involves communication at a higher level. This is something new, a device that has no set purpose and becomes a completely new invention every time you tell it what you expect of it. For this, you need something else new: the creation of a language. Not a human language, but something else: a language for communicating between human and machine. The computer has to be programmable.

The guests can be excused for not understanding the significance of this machine — a machine that, at this point, only exists in the eccentric mind of Charles Babbage. But Ada does. Over the next five years, she will take the lead in explaining this device to the world, and in demonstrating exactly how you could tell it to do things.

In the process, she will come to be known as the world's first computer programmer. Books and societies will be named after her, as well as a programming language. She will die at age 36.

Recent scholarship has confirmed the basic soundness of Babbage's designs. You can visit the Science Museum in London and see a working Difference Engine, finally built in the 1990s. And that's impressive. But the Analytical Engine is the real deal: bringing that design to life would prove that Charles Babbage effectively invented the computer, even if he couldn't get it built.
Which he couldn't. Much of the difficulty had to do with getting funding.

Dickens later satirized Babbage's unsuccessful battles with the British Treasury over support for the Analytical Engine in his novel *Little Dorrit.* The Babbage character is Daniel Doyce, "an inventor of mechanical devices, with a trail of successes in Paris and St Petersburg. He is back in England with vain hopes of getting a patent on a device he invented." Babbage fared no better.
And so Babbage's idea of a computer, a universal device for processing information, languished for a hundred years.

But that's another story.





## Written by Michael Swaine

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